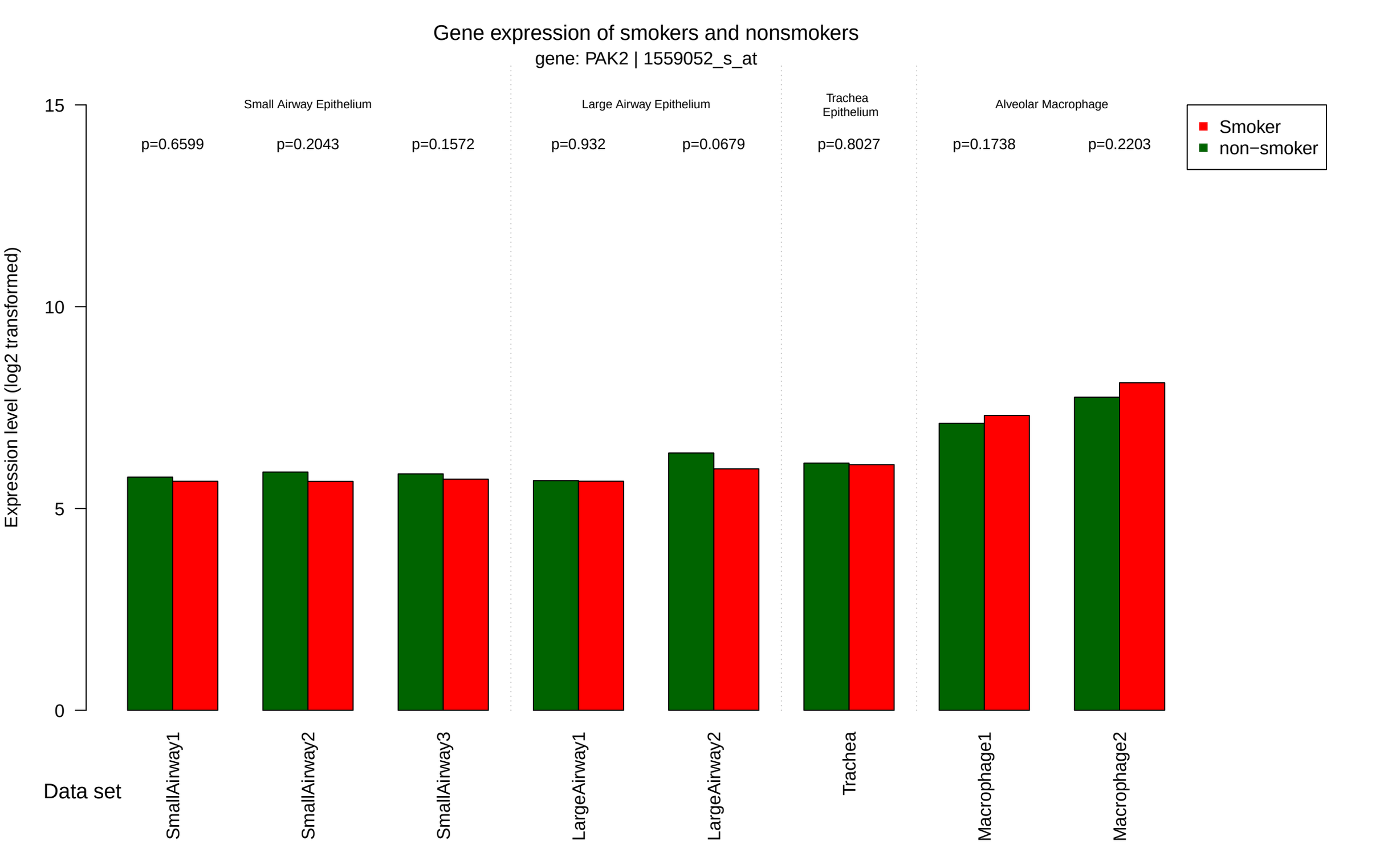
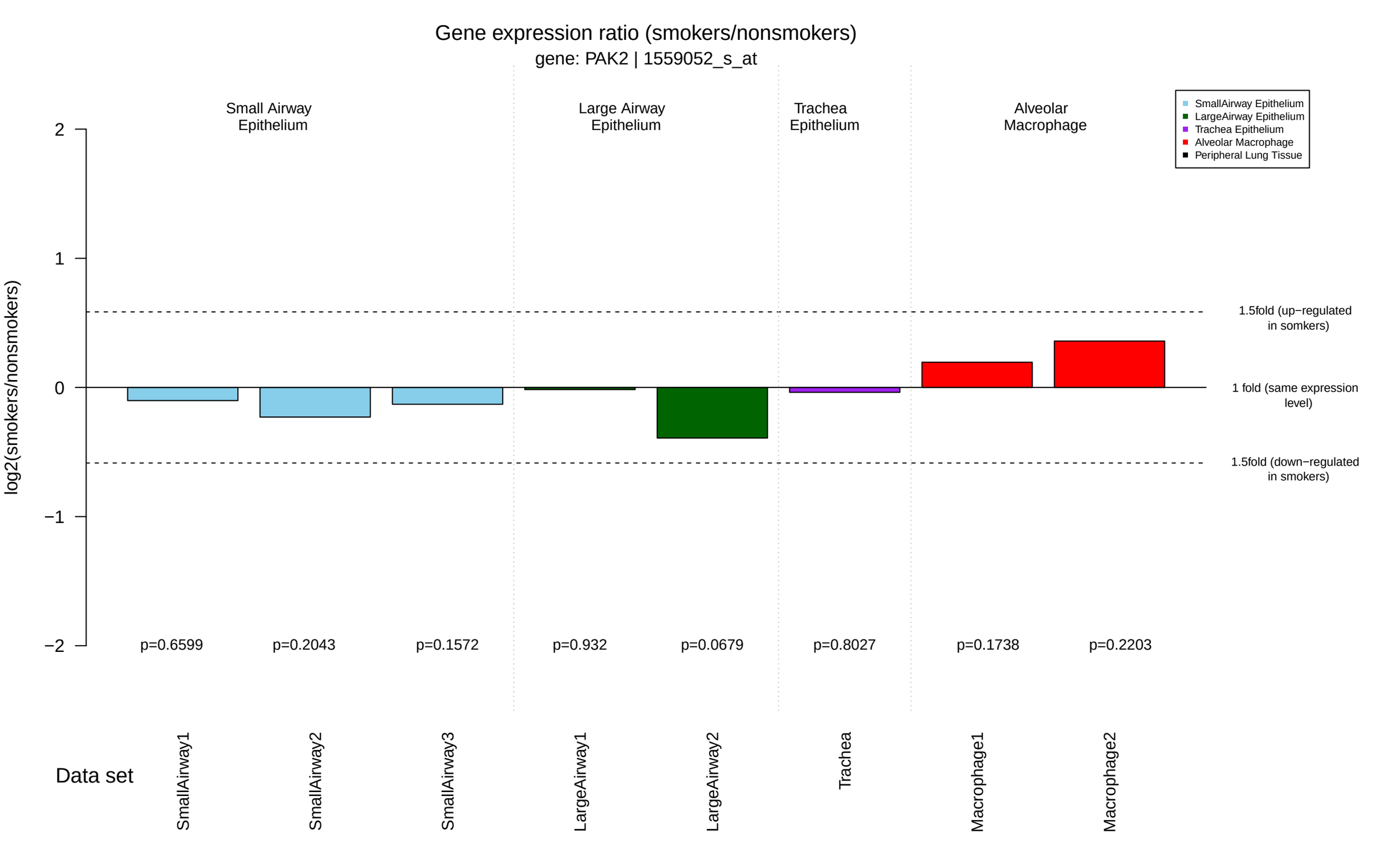
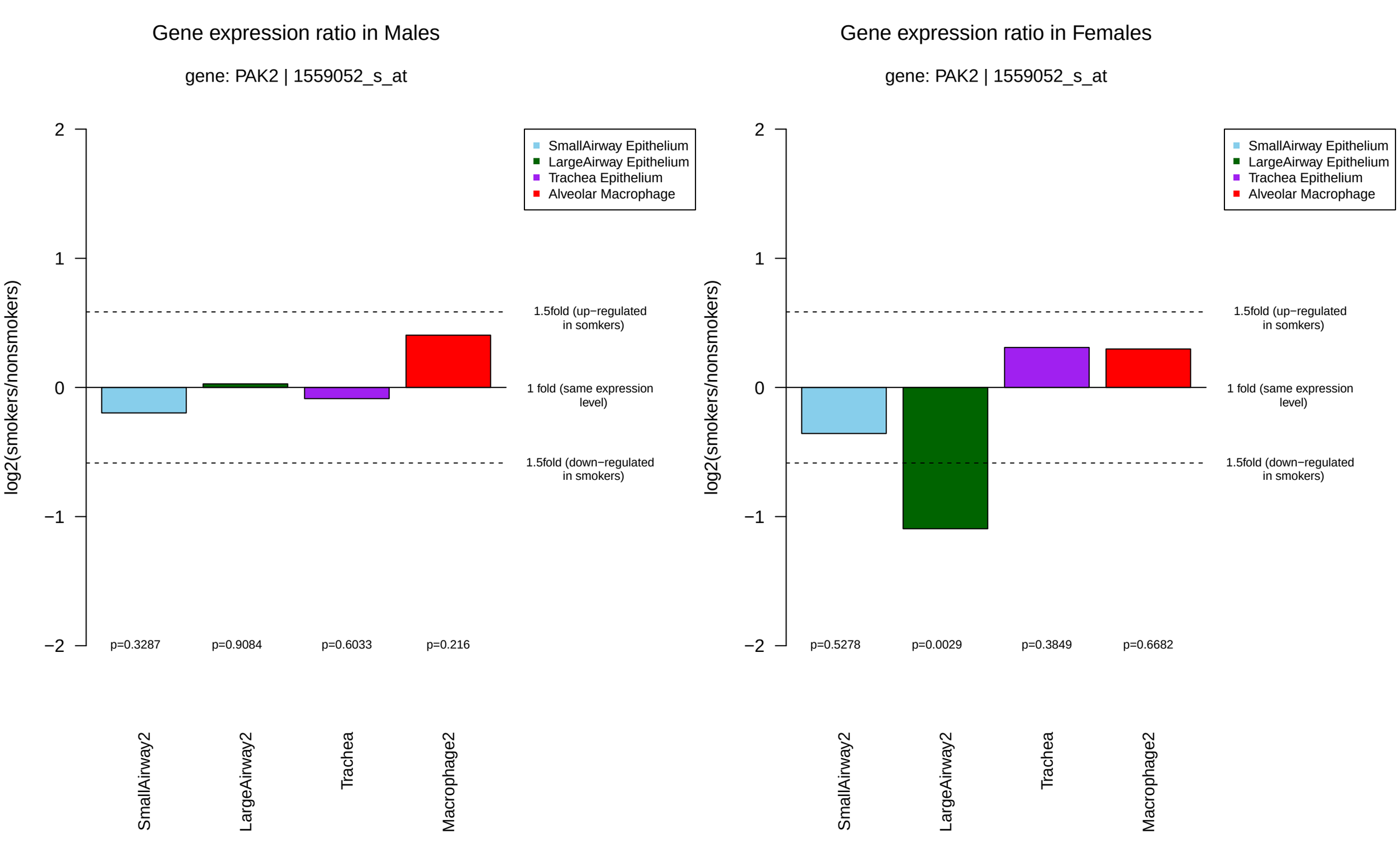
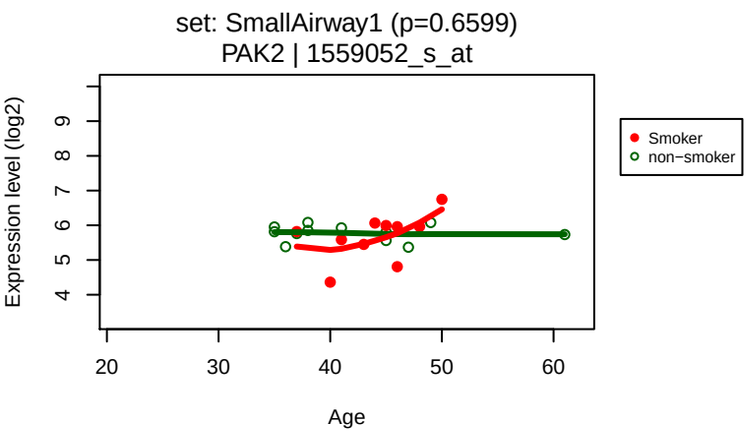
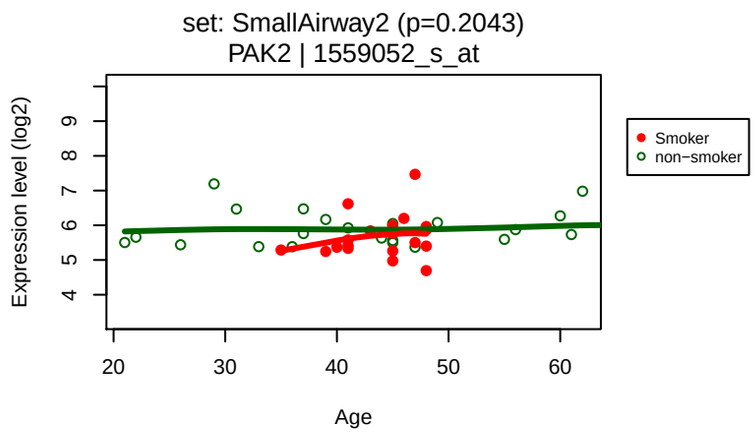
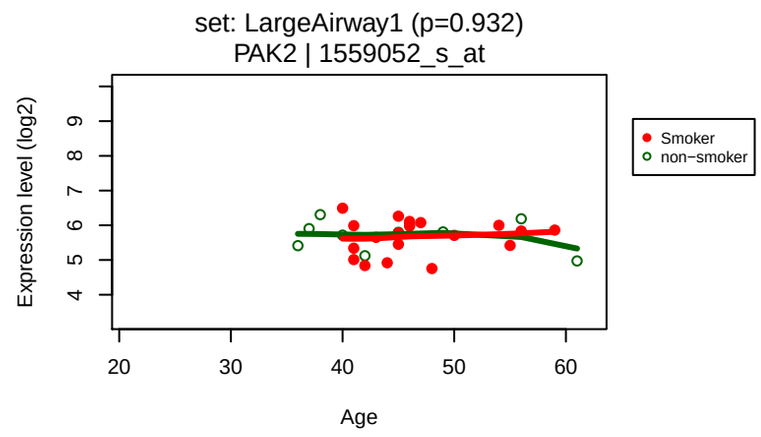

Gene expression of smokers and nonsmokers
gene: PAK2 | 1559052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6599
p=0.2043
p=0.1572
p=0.932
p=0.0679
p=0.8027
p=0.1738
p=0.2203
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

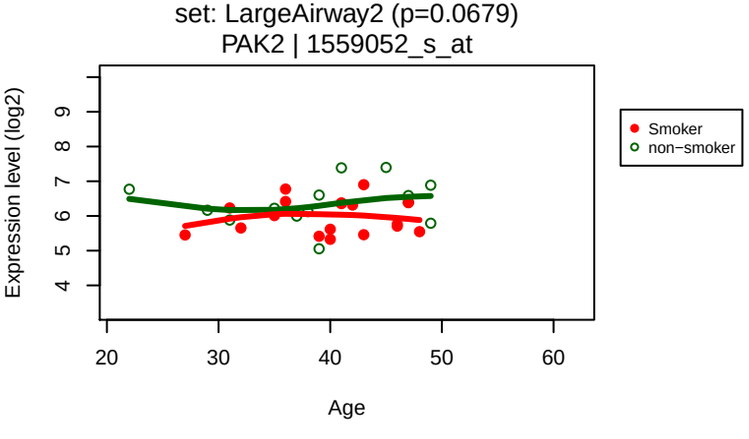

Gene expression ratio (smokers/nonsmokers)
gene: PAK2 | 1559052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6599
p=0.2043
p=0.1572
p=0.932
p=0.0679
p=0.8027
p=0.1738
p=0.2203
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

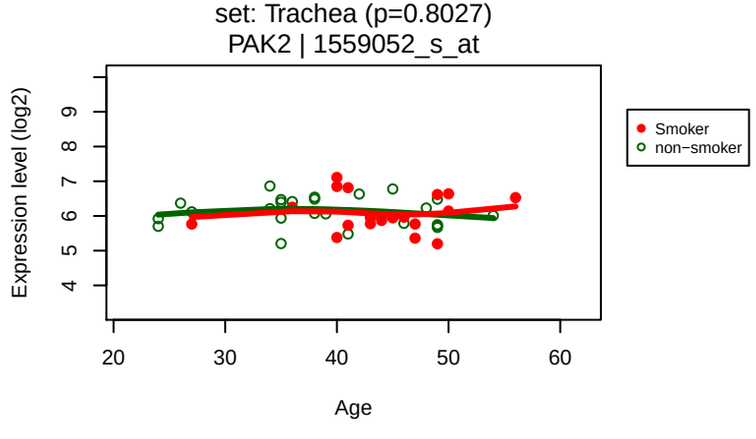

Gene expression ratio in Males
gene: PAK2 | 1559052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3287
p=0.9084
p=0.6033
p=0.216
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK2 | 1559052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5278
p=0.0029
p=0.3849
p=0.6682
SmallAirway2
LargeAirway2
Trachea
Macrophage2

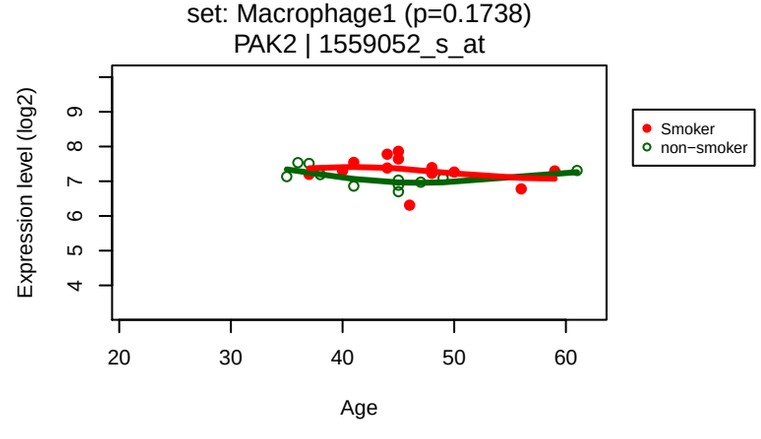

set: SmallAirway1 (p=0.6599)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker

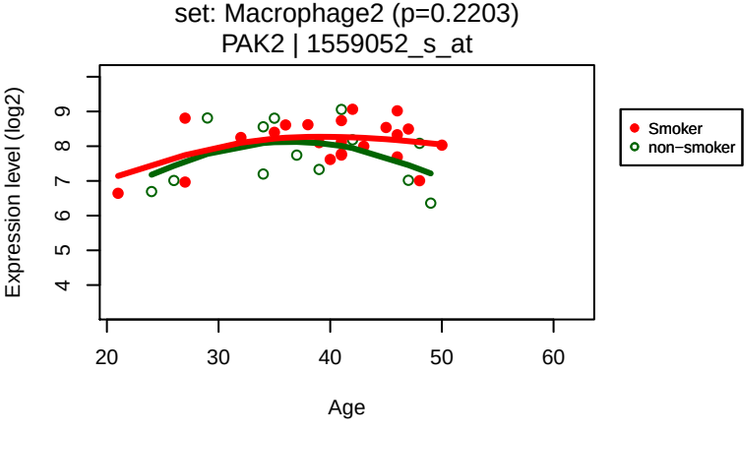

set: SmallAirway2 (p=0.2043)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker

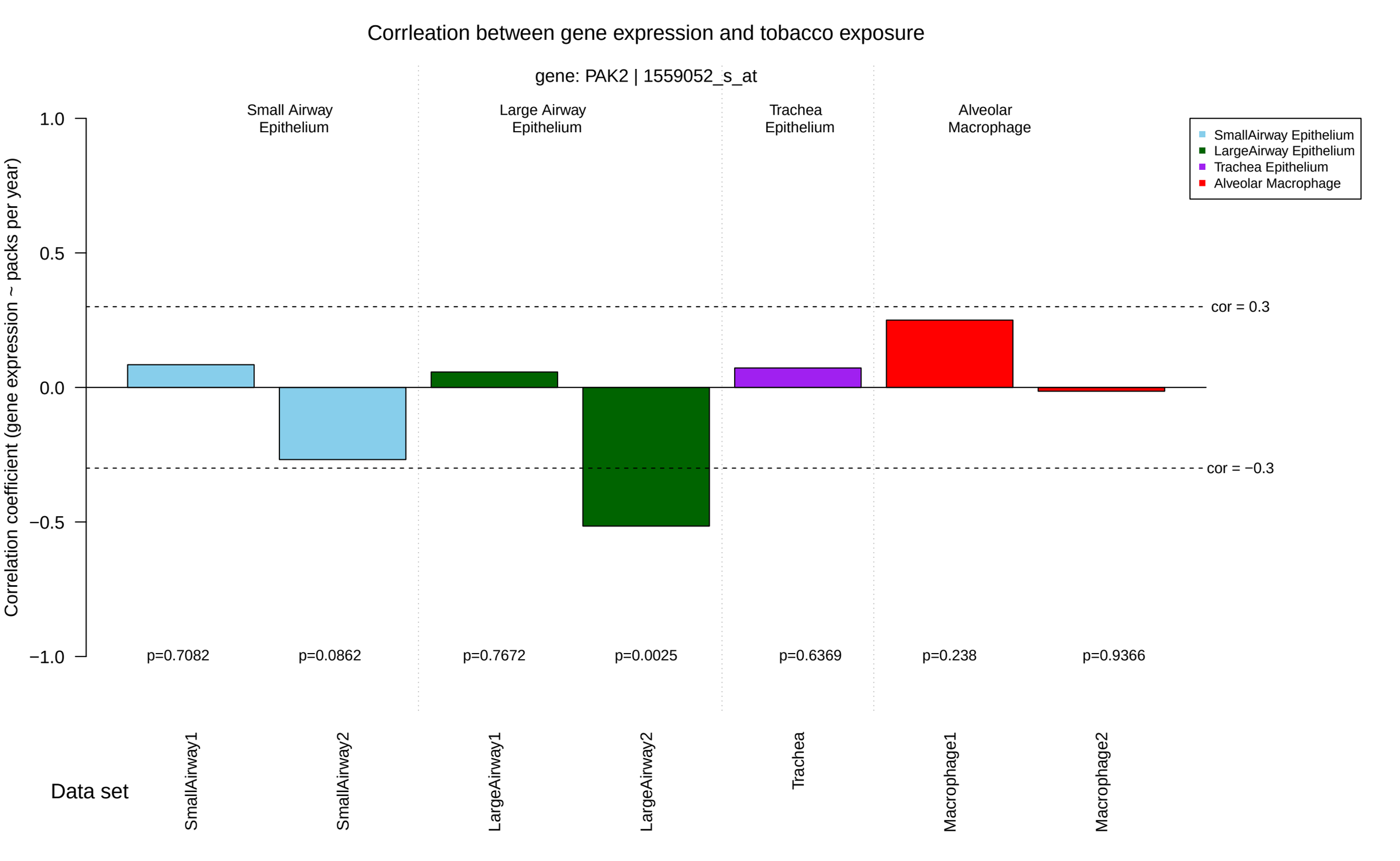

set: LargeAirway1 (p=0.932)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0679)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8027)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1738)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2203)
PAK2 | 1559052_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK2 | 1559052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7082
p=0.0862
p=0.7672
p=0.0025
p=0.6369
p=0.238
p=0.9366
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2